

An improved algorithm for arbitrary unitary synthesis

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Abstract

This paper presents an improved quantum gate decomposition method that results in more optimal quantum circuits. The current best-known exact method requires $O(\frac{22}{48}4^n)$ two-qubit gates for an arbitrary n -qubit unitary. We reduce this count to $O(\frac{21}{48}4^n)$ by introducing a parameterized single-qubit *pivot* gate that removes a two-qubit gate at each recursive step. Unlike conventional gradient-based approaches used to optimize gate parameters, we provide a principled method and show sufficient parameters always exist. We verify our decomposition with numerical experiments and discuss the tradeoffs and practicality.

I Introduction

I.A The synthesis problem

To run a quantum algorithm on hardware, its unitary operations must be translated into the native gate set of the device, a task known as gate decomposition or synthesis. The usual gate set is $\{\text{CX}\} \cup U(2)$, where CX is the two-qubit controlled-NOT. A parameter count shows that an exact decomposition of a generic n -qubit unitary needs at least [1]

$$c_{\min}(n) \geq \lceil \frac{1}{4}(4^n - 3n - 1) \rceil$$

CX gates. These gates are typically the bottleneck, since their connectivity constraints and error rates are a dominating factor over single-qubit gates.

I.B Existing work

The current approaches either provide an analytical decomposition or rely on numerical optimization. The best-known analytical methods [1, 2] follow the recursive pattern in Fig. 1. These have a direct algebraic formula but produce sub-optimal circuits, whereas numerical approaches can reach near-optimal circuits at the cost of intensive gradient-based search (see, e.g., [4, 5]). This work focuses on analytical decompositions which are more generalizable.

The quantum Shannon decomposition (QSD) [1] introduced the recursive pattern using three multiplexed rotations (“multiplexors”) between four sub-unitaries, requiring $O(\frac{23}{48}4^n)$ CX gates. The multiplexor [3] applies a different single-qubit rotation to a target qubit for each basis state of the control qubits. At each recursive step, it uses $D = 2^{n-1}$ CX gates for the bottom $n - 1$ control qubits and dominates the number of overall CX gates. The QSD was further improved using the Block-ZXZ decomposition [2] which modified the central multiplexor, resulting in the current best-known count $O(\frac{22}{48}4^n)$ for exact decomposition.

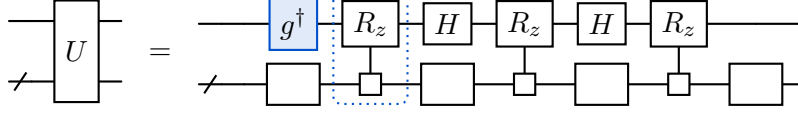


Figure 1: The template for decomposing a target n -qubit unitary U . We introduce a single-qubit pivot gate $g^\dagger$ (blue fill) which allows the left-side multiplexor (blue dots) to use one fewer CX gate by merging it into the central multiplexor. This template is applied recursively to the four generic $(n-1)$ -qubit unitaries (blank boxes), leading to a reduction of $(4^{n-2} - 1)/3$ CX gates.

I.C Our contribution

We adapt the current record-holding Block-ZXZ decomposition with a parameterized single-qubit gate at each recursive step, as depicted in Fig. 1 with blue highlights. This gate is chosen such that an additional CX gate can be merged into the central multiplexor, similar to the original optimizations used by Block-ZXZ to improve QSD. However, as outlined in Section II, the gate parameters do require iterative search, unlike the direct commutative optimizations used by Block-ZXZ. To remedy this, we provide a principled method to obtain these parameters which is unlike the conventional gradient-based approaches. Section III presents numerical experiments with a discussion of the practicality of our decomposition. Table 1 summarizes our main gate count result.

Number of qubits	1	2	3	4	5	6	n
QSD [1]	0	3	20	100	444	1868	$(23/48) \cdot 4^n - (3/2) \cdot 2^n + (4/3)$
Block-ZXZ [2]	0	3	19	95	423	1783	$(22/48) \cdot 4^n - (3/2) \cdot 2^n + (5/3)$
This work	0	3	18	90	402	1698	$(21/48) \cdot 4^n - (3/2) \cdot 2^n + 2$
Theoretical lower bound	0	3	14	61	252	1020	$(1/4) \cdot (4^n - 3n - 1)$

Table 1: CX counts for exact synthesis of a generic n -qubit unitary.

II An improved algorithm

Sec. II.A motivates the pivot gate, Sec. II.B states when the merge is possible, Sec. II.C shows how to compute the parameters, and finally Sec. II.D derives the main CX count result.

II.A Merging an additional CX into the central multiplexor

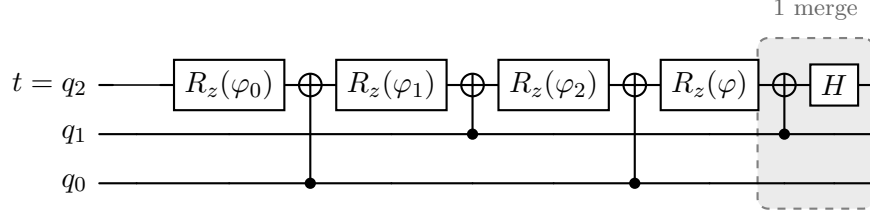
A generic multiplexed gate has a block-diagonal form $U_1 \oplus U_2$ with the well-known decomposition [1]:

$$U_1 \oplus U_2 = (I \otimes V) \Delta(\boldsymbol{\theta}) (I \otimes W), \quad U_1 U_2^\dagger = V \text{diag}(e^{-i\theta_j}) V^\dagger, \quad (1)$$

where the diagonal matrix $\Delta(\boldsymbol{\theta})$ is a multiplexed rotation, $\boldsymbol{\theta}$ are the eigenphases of $U_2 U_1^\dagger$, and V, W are unitary matrices. Here $j = (j_{n-2} \cdots j_0)_2 \in \{0, \dots, D-1\}$, with $D = 2^{n-1}$, labels the control basis states in binary, with bit j_i the state of q_i . This multiplexed rotation is depicted in Fig. 2a, where the single-qubit rotation angles are computed by [3]

$$\boldsymbol{\varphi} = \frac{1}{D} H_D \boldsymbol{\theta}, \quad (2)$$

(a) **Block-ZXZ: one CX merged from the left**



(b) **this work: two CX gates merged from the left**

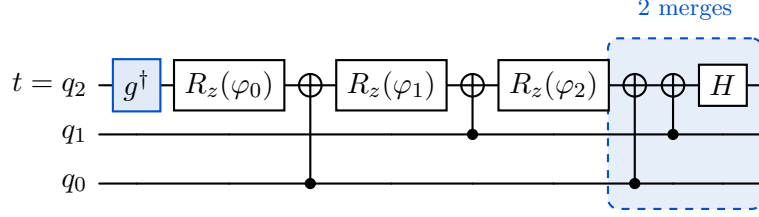


Figure 2: Detailed view of $g^\dagger$ and the left-side multiplexor (blue highlight of Fig. 1 for $n = 3$). (a) Block-ZXZ: $R_z(\varphi)$ separates the final CX gate from the Hadamard, so it cannot be merged into the central factor (grey). (b) This work: the pivot gate $g^\dagger$ makes $\varphi = 0$ so the CX gate can be merged. The central multiplexor (not shown) uses the same number of CX gates after merging, but now the left-side multiplexor uses 2 instead of 3.

where H_D is the $D \times D$ Walsh–Hadamard matrix with rows in Gray-code order. As shown in Fig. 2b, our improvement comes from removing the final angle,

$$\varphi = \frac{1}{D} \mathbf{h}^\top \boldsymbol{\theta}, \quad (3)$$

where $\mathbf{h}$ is the row of H_D corresponding to the top control qubit q_{n-2} , i.e., $h_j = (-1)^{j_{n-2}}$. For example, Fig. 2 has $n = 3$ so $j = (j_1 j_0)_2$ runs over $|q_1 q_0\rangle = |00\rangle, |01\rangle, |10\rangle, |11\rangle$, so $\mathbf{h} = (1, 1, -1, -1)$. Removing the final rotation with $\varphi = 0$ requires that the eigenphases $\boldsymbol{\theta}$ be balanced with $\mathbf{h}$. To achieve this, we define the gate $g = R_z(\alpha)R_y(\beta)$ and write the full decomposition as

$$U = (*) (I \oplus C(g)) (g^\dagger \otimes I), \quad (4)$$

where $I \oplus C(g)$ is the left-side multiplexor and $(*)$ collects the remaining right-side gates. We call g a *pivot* because it rotates the eigenspectrum of C to satisfy $\varphi = 0$. By (1), this modified left-side multiplexor can be decomposed as

$$C(g) = V \text{diag}(e^{i\theta_0}, \dots, e^{i\theta_{D-1}}) V^\dagger. \quad (5)$$

This decomposition is not unique. A common permutation of the eigenvalues and the columns of V leaves (5) unchanged, and each angle is fixed only modulo 2π , since a 2π shift flips the sign of one eigenvector, which V and W absorb. Writing $\mu_0, \dots, \mu_{D-1}$ for the eigenphases of $C(g)$, we have

$$\theta_j = \mu_{k(j)} + 2\pi m_j, \quad m_j \in \mathbb{Z}, \quad (6)$$

for any assignment $j \mapsto k(j)$ of eigenphases to basis states.

II.B When is the additional merge possible?

By (3), φ depends only on how the eigenphases are assigned to the +1 and -1 entries of $\mathbf{h}$, and by (6) we may choose this assignment. Let S , with $|S| = D/2$, be the set of eigenphases placed on the +1 entries. Then

$$\varphi = \frac{1}{D} \left(\sum_{k \in S} \mu_k - \sum_{k \notin S} \mu_k \right) + \frac{2\pi\ell}{D}, \quad \ell \in \mathbb{Z}, \quad (7)$$

where $\ell = \sum_{h_j=+1} m_j - \sum_{h_j=-1} m_j$ is set by the branches and can take any integer value. Hence φ can be set to zero iff

$$\sum_{k \in S} \mu_k - \sum_{k \notin S} \mu_k = -2\pi\ell \quad \text{for some } \ell \in \mathbb{Z}. \quad (8)$$

We call such an S a *balanced split*. For a fixed C this is nongeneric: the left side of (8) varies continuously with C , while the right side is discrete. The pivot, however, gives a continuous one-parameter family $C(\beta)$, so (8) becomes a scalar root condition in β .

II.C Choosing the pivot gate

We denote by α^* and β^* the pivot angles that produce a balanced split. The angle β moves C from $C(0)$ to $C(0)^{-1}$, while α is chosen so that an ordered eigenphase list reverses and changes sign between these endpoints.

The path $C(\beta)$. Write $U = \begin{pmatrix} X & Y \\ * & * \end{pmatrix}$ in $D \times D$ blocks, where $D = 2^{n-1}$ is divisible by four for $n \geq 3$, and assume X and Y are invertible, which holds for generic U . With $c = \cos(\beta/2)$ and $s = \sin(\beta/2)$, the pivot g gives $U(g \otimes I)$ the top-row blocks

$$X_g = e^{-i\alpha/2} c X + e^{i\alpha/2} s Y, \quad Y_g = -e^{-i\alpha/2} s X + e^{i\alpha/2} c Y. \quad (9)$$

At the endpoints the blocks swap with a sign change: (X_g, Y_g) is $(e^{-i\alpha/2} X, e^{i\alpha/2} Y)$ at $\beta = 0$ and $(e^{i\alpha/2} Y, -e^{-i\alpha/2} X)$ at $\beta = \pi$. For fixed α , the Block-ZXZ factorization [2, Sec. 4.1] gives the multiplexor matrix of (4) in terms of these pivoted blocks:

$$C(\beta) = -i X_g^\dagger (X_g X_g^\dagger)^{-1/2} (Y_g Y_g^\dagger)^{-1/2} Y_g, \quad (10)$$

where the inverse square roots are positive definite. The endpoint relation is

$$C(\pi) = C(0)^\dagger = C(0)^{-1}. \quad (11)$$

Thus the endpoint eigenphases are negatives of the starting ones, modulo 2π and up to permutation. We will use this symmetry to construct a continuous real phase imbalance with opposite endpoint values, forcing a zero at some $\beta^* \in [0, \pi]$.

Choosing α . In (10) the inverse square roots have positive determinants and $(-i)^D = 1$, so $\det C$ has the phase of $\det Y_g / \det X_g$. With τ_k the eigenvalues of $e^{i\alpha} X^{-1} Y$, (9) gives

$$\frac{\det Y_g}{\det X_g} = \prod_k \left(\tau_k \frac{c - s/\tau_k}{c + s\tau_k} \right). \quad (12)$$

As β runs from 0 to π , the factor $c + s\tau_k$ moves from 1 to τ_k and $c - s/\tau_k$ from 1 to $-1/\tau_k$. For nonreal τ_k and $\beta > 0$, both lie in the open half-plane containing τ_k and cannot vanish. Thus

choosing α with no real τ_k keeps X_g, Y_g invertible and $C(\beta)$ continuous. The two factors accumulate phases $\arg \tau_k$ and $\text{sgn}(\arg \tau_k) \pi - \arg \tau_k$, respectively. Subtracting the denominator phase from the numerator phase gives the total change in the continuous phase of $\det C$:

$$\pi(N_+ - N_-) - 2 \sum_k \arg \tau_k, \quad \arg \tau_k \in (-\pi, \pi), \quad (13)$$

where $N_{\pm}$ count the τ_k in the upper and lower half-planes. Choosing α^* with $N_+ = N_- = D/2$ removes the half-turn term. Varying α rotates all τ_k together. With distinct crossing angles, N_+ changes by one at each real-axis crossing; a rotation by π replaces N_+ by $D - N_+$. The count therefore reaches $D/2$ on an interval between crossings, and we choose α^* in its interior.

Choosing β . With $\alpha = \alpha^*$, let $\sigma(\beta)$ be the continuous phase of $\det C(\beta)$ with $\sigma(0) = \sum_k \arg \tau_k$. By (13),

$$\sigma(\pi) = -\sigma(0). \quad (14)$$

Choose real representatives of the eigenphases of $C(\beta)$, ordered by rank k :

$$\nu_0(\beta) \leq \dots \leq \nu_{D-1}(\beta) \leq \nu_0(\beta) + 2\pi, \quad \sum_k \nu_k(\beta) = \sigma(\beta). \quad (15)$$

The interval condition fixes the ordered list up to cyclic reordering with 2π branch shifts. Moving the smallest phase to the end after adding 2π increases the sum by 2π , so the sum condition selects a unique list. This choice varies continuously with β , including when a phase crosses the argument cut. Negating an increasing list reverses its order, so (11) and (14) give $\nu_k(\pi) = -\nu_{D-1-k}(0)$. The middle half $S = \{D/4, \dots, 3D/4 - 1\}$ of the ranks is invariant under $k \mapsto D - 1 - k$, so the imbalance

$$P(\beta) = \sum_{k \in S} \nu_k(\beta) - \sum_{k \notin S} \nu_k(\beta) \quad (16)$$

satisfies

$$P(\pi) = -P(0), \quad (17)$$

and by continuity $P(\beta^*) = 0$ for some $\beta^* \in [0, \pi]$. There the sums over the two halves agree. Taking $\mu_k = \nu_k(\beta^*)$, the middle half S satisfies (8) with $\ell = 0$. Placing these lifted eigenphases on the basis states with $h_j = +1$ and the rest on $h_j = -1$, with no additional branch shifts ($m_j = 0$ in (6)), gives $\varphi = 0$, and the pivot $g = R_z(\alpha^*)R_y(\beta^*)$ lets the additional CX merge.

II.D Gate count

Block-ZXZ needs at most $c_n \leq 4c_{n-1} + 3 \cdot 2^{n-1} - 5$ CX gates [2, Sec. 5.3]: four $(n-1)$ -qubit children and three multiplexors of 2^{n-1} CX gates each, less one CX for each of the three children decomposed up to a diagonal and one for each of the two CX gates merged into the central multiplexor. The pivot merges a third, so

$$c_n = 4c_{n-1} + 3 \cdot 2^{n-1} - 6, \quad c_2 = 3. \quad (18)$$

Unrolling over the $(4^{n-2} - 1)/3$ recursion nodes as in [2, Sec. 5.3],

$$c_n = 4^{n-2}c_2 + 3 \cdot 2^{n-1}(2^{n-2} - 1) - 2(4^{n-2} - 1) = \frac{21}{48}4^n - \frac{3}{2}2^n + 2.$$

Subtracting (18) from the Block-ZXZ recurrence, the saving $\Delta_n = c_n^{\text{ZXZ}} - c_n$ satisfies $\Delta_n = 4\Delta_{n-1} + 1$ with $\Delta_2 = 0$, so

$$\Delta_n = \sum_{k=0}^{n-3} 4^k = \frac{4^{n-2} - 1}{3}, \quad (19)$$

one CX for each recursion node. This agrees with the difference of the closed forms, $(\frac{22}{48} - \frac{21}{48})4^n + \frac{5}{3} - 2 = \frac{4^{n-2}-1}{3}$.

III Results and Discussion

We implemented the improved decomposition of Sec. II in Python using standard linear algebra primitives provided with numpy and scipy.

Figure 3 compares the CX counts and total synthesis times for our proposed decomposition against the current best-known Block-ZXZ decomposition. These comparisons use the same random unitary instances. The target unitary matrix and reconstructed circuit matrix are equal to machine precision. All experiments were run on a MacBook Pro (M1 series, 2021).

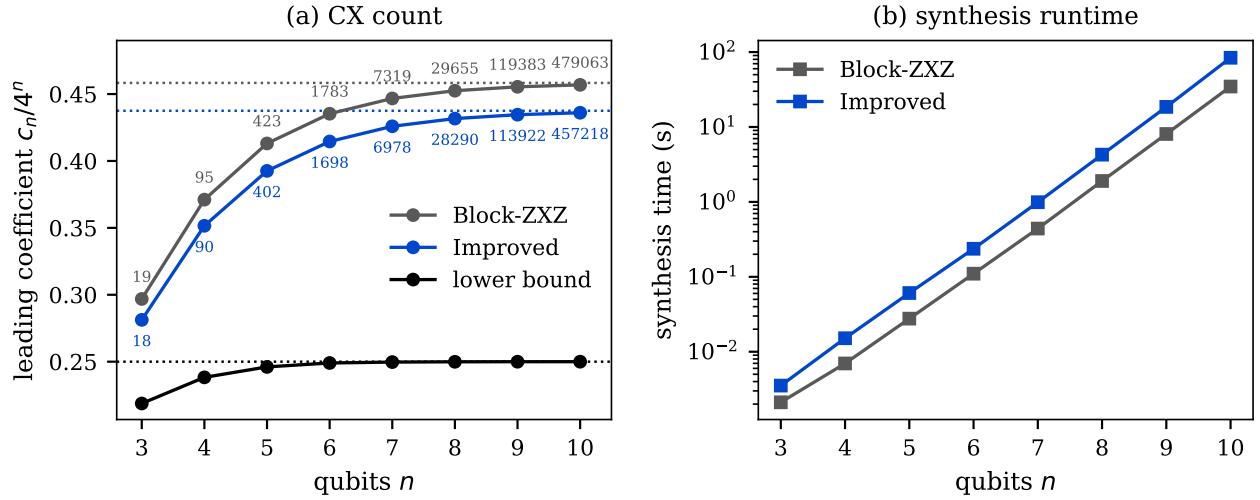


Figure 3: Our improved decomposition (blue) against the best-known baseline Block-ZXZ (grey). (a) CX count divided by 4^n with the lower bound in black. The dotted lines mark asymptotic coefficients $22/48$, $21/48$, and $12/48$. (b) Total synthesis time on a logarithmic axis.

Our decomposition method requires $\frac{4^{n-2}-1}{3}$ fewer CX gates while incurring roughly $2\times$ added runtime cost, which appears to be constant.

Each recursive step requires a one-dimensional root search with a guaranteed solution. The CX count is independent of the method used to locate the root, so perhaps further runtime improvements can be made.

A second pivot applied in reverse to the right-side multiplexor could perhaps save another $(4^{n-2} - 1)/3$ CX gates, though the two pivots are coupled and would require a joint search.

References

- [1] V. V. Shende, S. S. Bullock, I. L. Markov, *Synthesis of quantum logic circuits*, IEEE TCAD 25, 1000 (2006), arXiv:quant-ph/0406176.
- [2] A. M. Krol, Z. Al-Ars, *Beyond quantum Shannon decomposition: circuit construction for n -qubit gates based on block-ZXZ decomposition*, Phys. Rev. Applied 22, 034019 (2024), arXiv:2403.13692.

- [3] M. Möttönen, J. J. Vartiainen, V. Bergholm, M. M. Salomaa, *Quantum circuits for general multiqubit gates*, Phys. Rev. Lett. 93, 130502 (2004), arXiv:quant-ph/0404089.
- [4] D. Wierichs, J. S. Kottmann, N. Killoran, *Unitary synthesis with optimal brick wall circuits*, arXiv:2511.16736.
- [5] P. Rakyta, Z. Zimborás, *Approaching the theoretical limit in quantum gate decomposition*, Quantum 6, 710 (2022).